

SGA 7-I. Exposé IX. Néron Models and Monodromy

Conclusion

13.5. The relatively algebraic situation

We now suppose that the analytic group A comes from a relative group scheme over S , smooth and separated, whose fibres are extensions of abelian schemes by tori. We denote this relative scheme again by A . This abuse of notation is justified by the fact that the functor

$$G \longmapsto G^{\text{an}}$$

from relative group schemes of this type to analytic groups over S is fully faithful. This is a relative GAGA-type assertion. For the notion of a relative scheme, and for the relation between relative schemes over S and analytic spaces over S , the references are Hakim's book and the corresponding earlier exposés on relative analytic geometry.

Let O be the local ring of S at s . Near s , the relative scheme A is determined by the ordinary scheme A_O over $\text{Spec } O$ obtained by localizing A at s . The formal completion of A_O along its special fibre, which is isomorphic to $A_s = A_0$, is the same as the formal completion of the analytic group A along A_s , after identifying

$$A_n = A \times_S S_n$$

with algebraic schemes over S_n . In the favourable case where this formal group scheme, viewed over the completed local ring $\hat{O}$, is algebraizable, one obtains a group scheme over $\text{Spec } \hat{O}$, extension of an abelian scheme $B_{\hat{O}}$ by a torus. To simplify the statement, assume that $B_{\hat{O}}$ is projective over $\hat{O}$; this is the case, for instance, when S is normal at s .

Proposition 13.5.1. *Under the preceding hypotheses, the analytic group $A^{\natural}$ of (13.3.3) comes, in a neighbourhood of s , from a relative group scheme determined up to a unique isomorphism. This group scheme is an extension of a projective relative abelian scheme over S by a torus. If $A_{\hat{O}}^{\natural}$ denotes the group scheme over $\hat{O} = \hat{O}_{S,s}$ that it defines, then $A_{\hat{O}}^{\natural}$, with its extension structure, is canonically isomorphic to the Raynaud extension introduced above.*

The last assertion follows at once from the first and from 7.2.1, since $A_{\hat{O}}^{\natural}$ has, along its special fibre, the formal completion prescribed by (13.3.6). To prove algebraicity of $A^{\natural}$ near s , one proceeds as in 7.2.1 and is reduced immediately to proving algebraicity of the abelian quotient B of $A^{\natural}$.

Lemma 13.5.2. *Let B be an analytic abeloid group over the analytic space S . Let $s \in S$ be a point such that the formal completion $\hat{B}$ along the fibre B_s is algebraizable. Thus the spaces $B_n = B \times_S S_n$ are algebraizable, and $\hat{B}$, viewed as a formal scheme over $\hat{O}_{S,s}$, comes from an abelian scheme $B_{\hat{O}}$ over $\hat{O}$. Suppose moreover that $B_{\hat{O}}$ is projective over $\hat{O}$. Then B is projective over S after replacing S by a neighbourhood of s .*

The hypothesis means that the formal analytic space $\hat{B}$ has a polarization, that is, a compatible system of polarizations p_n on the B_n . The conclusion means that there is a polarization p of B over a neighbourhood of s . Let P be the analytic space over S of relative polarizations of B . It represents the functor which assigns to $S' \rightarrow S$ the set of polarizations of $B_{S'}$ over S' . This is an open subfunctor of the relative Néron-Severi functor $\text{NS}_{B/S}$. Standard arguments show that $\text{NS}_{B/S}$, hence also P , is represented by an unramified analytic space over S . The formal polarization gives a formal section of P at s . Since P is unramified and locally quasi-finite over S , a formal section is induced by an analytic section over a neighbourhood of s . This analytic section is the desired polarization of B .

Remarks 13.5.3. (a) If the projectivity hypothesis on $B_{\hat{O}}$ is omitted, one can still conclude that B is algebraic near s . One uses an unpublished theorem, independently due to Artin and Deligne, saying that if X is proper over S and if the formal completion of X along X_s is algebraizable, then X itself is algebraizable. This reduces matters to the case where S is reduced. When S is normal, an abelian scheme over $\hat{O}$ is projective, and Lemma 13.5.2 applies. The general reduced case is obtained by non-flat descent from the normalization of S , using the fact that a morphism for which $\mathcal{O}_S \rightarrow f_*\mathcal{O}_{S'}$ is injective is an effective descent morphism for relative abelian schemes.

(b) The same method should also prove the analogous result for any analytic space B proper over S . The more delicate question is whether an analytic space X proper over S , whose formal completion along X_s is algebraizable, is itself algebraizable near s . Here algebraizability must be understood in a wider sense than in Hakim's book: near s , the space should come from an algebraic space over the local base, in Artin's sense.

Still under the hypotheses of Proposition 13.5.1, choose for every $n > 0$ the standard isomorphism

$$\mu_n \simeq (\mathbb{Z}/n\mathbb{Z})_S, \quad m \longmapsto \exp(2\pi i m/n),$$

and, by passing to inverse limits over powers of a fixed prime ℓ , identify the usual Tate twist with the constant $\mathbb{Z}_\ell$ -system. Applying the exact sequence of $\text{Ext}^1(\mathbb{Z}/n\mathbb{Z}, -)$ to the exact sequence (13.1.3) gives the corresponding canonical projective systems. In the present algebraic situation the finite sheaves that occur are represented by relative finite group schemes, and the analytic symbols used earlier therefore acquire an algebraic meaning.

The resulting identifications are compatible with fixed parts and toric parts. If A' satisfies the same hypotheses and if W is a biextension of (A, A') by $\mathbb{G}_m$, defining a duality over the open set where A and A' are abelian schemes, then the induced pairings on Tate modules agree with the algebraic pairings. After shrinking around s , the ℓ -adic pairings glue to an integral pairing

$$R \times R' \longrightarrow \mathbb{Z}. \tag{13.5.11}$$

The algebraic orthogonality theorem can then be applied to R . The pairing (13.5.11) is zero on the fixed and toric products prescribed by the algebraic theory. If S is nonsingular of dimension one and if, over $U = S - \{s\}$, the restrictions of A and A' are dual abelian

schemes, the fixed and toric subobjects are mutual annihilators near s . Since (13.5.11) is a perfect duality over U , it follows that R/R^t is locally constant and that the local monodromy action on R is unipotent of exponent at most two. One also obtains a perfect duality between R/R^f and the dual toric lattice. Under this identification the homomorphism of (13.3.4) may be written

$$i : M \longrightarrow A^\natural. \quad (13.5.13)$$

This is the form in which it will reappear below.

14. Links with transcendental theory: the rigid-analytic case

14.1. The canonical homomorphism $M_K \rightarrow G_K$

We assume the foundations of rigid-analytic spaces. Let S be a complete trait and let A_K be an abelian scheme over its fraction field K . Section 8.1 constructs, by descent from the semi-stable case and from the Raynaud extension, an extension

$$0 \longrightarrow T_K \longrightarrow G_K \longrightarrow B_K \longrightarrow 0 \quad (14.1.1)$$

of an abelian scheme B_K by a torus T_K . For the dual abelian scheme A'_K one has similarly

$$0 \longrightarrow T'_K \longrightarrow G'_K \longrightarrow B'_K \longrightarrow 0, \quad (14.1.2)$$

where B'_K is dual to B_K . Let

$$M_K = D(T'_K), \quad M'_K = D(T_K) \quad (14.1.3)$$

be the character lattices. By Exposé VIII, the two extension structures are equivalently described by homomorphisms

$$M_K \xrightarrow{p_K} B'_K, \quad M'_K \xrightarrow{p'_K} B_K. \quad (14.1.4)$$

Let W_K be the biextension of (B_K, B'_K) by $\mathbb{G}_m$ that gives the canonical duality, and let E_K be its inverse image on (M_K, M'_K) . A trivialization of E_K is canonically the same as a homomorphism

$$i : M_K \longrightarrow G_K \quad (14.1.5)$$

lifting p_K , or, symmetrically, as a homomorphism

$$i' : M'_K \longrightarrow G'_K \quad (14.1.6)$$

lifting p'_K .

Writing a superscript an for the associated rigid-analytic groups, Raynaud defines a

canonical exact sequence

$$0 \longrightarrow M_K^{\text{an}} \xrightarrow{i^{\text{an}}} G_K^{\text{an}} \xrightarrow{e} A_K^{\text{an}} \longrightarrow 0. \quad (14.1.7)$$

The homomorphism i^{an} can also be read as an ordinary homomorphism $i : M_K \rightarrow G_K$, lifting p_K . The sequence is the rigid-analytic analogue of (13.3.4), functorial in A_K and compatible with change of traits.

By this compatibility one reduces to the semi-stable case. There the groups $T_K, G_K, B_K, M_K, \dots$ are generic fibres of group schemes $T, G, B, M, \dots$ over S , and G is an extension of B by T . Let $\widehat{G}$ be the formal completion of G along the special fibre. There is a canonical isomorphism

$$\widehat{G} \xrightarrow{\sim} \widehat{A}^0, \quad (14.1.8)$$

and, on rigid generic fibres,

$$\widehat{G}^{\text{an}} \xrightarrow{\sim} \widehat{A}^{0,\text{an}}. \quad (14.1.9)$$

There are also canonical open immersions

$$\widehat{G}^{\text{an}} \hookrightarrow G_K^{\text{an}}, \quad \widehat{A}^{0,\text{an}} \hookrightarrow A_K^{\text{an}}. \quad (14.1.10)$$

The map e is characterized uniquely by the commutative square

$$\begin{array}{ccc} \widehat{G}^{\text{an}} & \longrightarrow & \widehat{A}^{0,\text{an}} \\ \downarrow & & \downarrow \\ G_K^{\text{an}} & \xrightarrow{e} & A_K^{\text{an}}, \end{array} \quad (14.1.11)$$

where the unnamed arrows are those of (14.1.9) and (14.1.10).

Remarks 14.1.12. (a) The uniqueness of e is immediate from connectedness of $\widehat{G}^{\text{an}}$; existence is the non-trivial part and is supplied by Raynaud's rigid-analytic theory. The definition of i^{an} is likewise transcendental, although once G is constructed by formal geometry the resulting i is an algebraic datum of the form (14.1.5). At this stage no purely algebraic-geometric description is known. The arithmetic characterization below is obtained by reduction to a situation of finite type over $\mathbb{Z}$.

(b) The homomorphism i is canonically associated with a determined trivialization of E_K , or equivalently with the dual homomorphism i' .

14.2. Relation with monodromy

Let $n > 0$. Applying the exact sequence of $\text{Ext}^1(\mathbb{Z}/n\mathbb{Z}, -)$ to (14.1.7) on the rigid-analytic site over K , and using that M_K^{an} is torsion-free and G_K^{an} is n -divisible, gives an exact sequence of finite group schemes

$$0 \longrightarrow G_K[n] \longrightarrow A_K[n] \longrightarrow M_K/nM_K \longrightarrow 0. \quad (14.2.1)$$

Passing to inverse limits over powers of a prime ℓ yields

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow M_K \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.2.2)$$

The purely algebraic formulas of 8.1.2 and 8.1.7 give the same exact sequence:

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow T_\ell(A_K)/T_\ell(A_K)^f \simeq M_K \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.2.3)$$

The compatibility follows from (14.1.11). This compatibility, for a fixed ℓ , already characterizes i transcendently: the map e identifies the quotients $T_\ell(A_K)/T_\ell(A_K)^f$ with the ℓ -adic completions of a single lattice $\text{Ker } e$, compatibly with the integral structure coming from the algebraic theory. Thus

$$i : M_K \xrightarrow{\sim} \text{Ker } e. \quad (14.2.3 \text{ bis})$$

The same homomorphism reconstructs the monodromy pairing

$$u : M \times M' \longrightarrow \mathbb{Z}. \quad (14.2.4)$$

In the semi-stable case, assume for simplicity that M and M' are constant lattices. Pulling the Poincaré biextension on the abelian quotients back by the two homomorphisms from M and M' gives a biextension E of (M, M') by $\mathbb{G}_m$. For $(m, m') \in M \times M'$, the fibre $E_{m, m'}$ corresponds to an invertible V -module. Choosing a basis defines, independently of the choice, a valuation map

$$v : E_{m, m'}(K) \longrightarrow \mathbb{Z}.$$

A trivialization φ of E gives

$$(m, m') \longmapsto v(\varphi(m, m')), \quad (14.2.5)$$

and the monodromy homomorphism is this pairing for the splitting attached to i .

Remarks 14.2.6. Thus the rigid-analytic theory recovers integrality and independence of ℓ of the monodromy pairing. Raynaud's theory also recovers its positivity.

Remark 14.2.7. Regard $M \otimes \mathbb{Q}$ as an extension of $(M \otimes \mathbb{Q})/M$ by M . Using $i : M \rightarrow G_K$ gives

$$0 \longrightarrow G_K \longrightarrow G_K^\natural \longrightarrow (M \otimes \mathbb{Q})/M \longrightarrow 0. \quad (14.2.8)$$

Since G_K is divisible, this is equivalent to the family of extensions (14.2.2). It is therefore defined by algebraic data and is canonically the group introduced in (2.2.3), after identifying its component group with $(M \otimes \mathbb{Q})/M$.

14.3. An arithmetic characterization of i

For every prime ℓ , the extension (14.2.2) is known algebraically. Hence one knows the image of i under

$$\mathrm{Hom}(M_K, G_K) \longrightarrow \prod_{\ell} \mathrm{Ext}^1(M_K \otimes \mathbb{Z}_{\ell}, T_{\ell}(G_K)). \quad (14.3.1)$$

When M_K is constant with value M , this is the homomorphism

$$\mathrm{Hom}(M, G_K(K)) \longrightarrow \prod_{\ell} \mathrm{Hom}(M, H^1(K, T_{\ell}(G_K))) \quad (14.3.2)$$

obtained from the Kummer boundary maps

$$G_K(K) \longrightarrow \prod_{\ell} H^1(K, T_{\ell}(G_K)). \quad (14.3.3)$$

The kernel consists of those elements of $G_K(K)$ divisible by every integer $n > 1$.

Lemma 14.3.4. *Let V be a discrete valuation ring with residue field k of characteristic $p > 0$, of finite type over the prime field. Let G_K be a finite type group scheme over K , with zero unipotent rank if $\mathrm{char} K = 0$. Then the subgroup of infinitely divisible elements of $G_K(K)$ is zero. Equivalently, when G_K is divisible, (14.3.3) is injective.*

Proof. After a finite extension of K and normalization of V , reduce to the case where G_K has a composition series with quotients among finite constant groups, abelian varieties, $\mathbb{G}_m$, and additive groups. The finite constant and multiplicative cases are immediate. For an abelian variety, pass to the Néron model G and reduce to proving that the infinitely divisible subgroup of $G(S)$ is zero. This is true on the special fibre by the usual dévissage and the Mordell-Weil theorem over a field of finite type. The kernel of $G(S) \rightarrow G(k)$ is filtered by vector groups over k , killed by p on successive quotients; intersection over all infinitesimal levels is zero. The additive case is handled similarly.

Remark 14.3.4.1. It is essential to require divisibility by all integers, not only by powers of one prime. For a smooth group over a henselian trait, elements in the kernel of reduction are infinitely ℓ -divisible for every $\ell \neq p$. The preceding argument shows that the infinitely p -divisible subgroup is finite, and the full intersection is zero.

When k has characteristic $p > 0$ and is of finite type in the absolute sense, the homomorphism i is therefore determined by its image under (14.3.1), equivalently by the family of extensions (14.2.2). To obtain a general characterization, let $S = \mathrm{Spec} V$, with V a normal noetherian ring, separated and complete for an adic topology. Let A be a smooth quasi-projective group scheme over S , with connected fibres, such that the special fibre is an extension of an abelian scheme by an isotrivial torus and such that A_U is an abelian scheme over a non-empty open $U \subset S - S_0$. Section 7 constructs a group scheme G over S , an extension of an abelian scheme B by a torus T , characterized by an isomorphism of

formal completions

$$\widehat{G} \xrightarrow{\sim} \widehat{A}^0. \quad (14.3.5.1)$$

For every prime ℓ one obtains on U an exact sequence

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow M \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.3.5.3)$$

If A' satisfies the same hypotheses and W is a biextension of (A, A') by $\mathbb{G}_m$, inducing a duality over U , the resulting pairings on Tate modules give the orthogonality theorem and hence the identification

$$T_\ell(A_U)/T_\ell(A_U)^f \simeq M \otimes \mathbb{Z}_\ell. \quad (14.3.5.5)$$

Thus one obtains the extension

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow M \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.3.5.6)$$

Conjecture 14.4. *In the situation described in 14.3.5, there exists a homomorphism*

$$i : M_U \longrightarrow G_U \quad (14.4.1)$$

such that, for every prime ℓ , the extension (14.3.5.6) is isomorphic to the extension deduced from i as in (14.2.2). More precisely, there is one and only one way to assign such an i to every situation of the type of 14.3.5, compatibly with all base changes $S' \rightarrow S$ arising from continuous homomorphisms $V \rightarrow V'$ of noetherian adic rings.

The existence of i , compatibly with base change, should follow from Raynaud's rigid-analytic theory in its general adic form. For uniqueness, one reduces to the case where S_0 is of finite type over $\mathbb{Z}$. Indeed, V is a filtered inductive limit of normal subrings of finite type over $\mathbb{Z}$, and the situation descends, for large index, to such a subring or to its separated completion for the induced adic topology. Since $S - S_0$ is Jacobson, two possible homomorphisms are equal if they agree at all closed points of U . Normalizing the closure of such a point gives a complete trait with finite residue field, and the uniqueness result of the preceding one-dimensional case applies. This proves uniqueness, and then base-change compatibility follows from uniqueness, assuming existence.

Remark 14.5. It remains to formulate and prove the expected compatibility between the rigid-analytic homomorphism i and the analogous complex-analytic homomorphism (13.5.13), or better, to find a common generalization of both constructions.

Bibliography

- [1] M. Artin, *Algebraization of formal moduli I*, to appear.
- [2] H. Cartan, *Familles d'espaces complexes et fondements de la géométrie analytique*, Séminaire ENS 1966/67.

- [3] P. Deligne and D. Mumford, The irreducibility of the space of curves of given genus, *Publ. Math. IHES* 36 (1969).
- [4] P. Deligne, work in preparation on Hodge theory, to appear in *Publ. Math. IHES*.
- [5] M. Demazure, J. Giraud, and M. Raynaud, *Schémas abéliens*, Séminaire Orsay 1967/68, to appear.
- [6] A. Grothendieck, Le groupe de Brauer III, in *Dix exposés sur la cohomologie des schémas*, North-Holland.
- [7] A. Grothendieck, Crystals and the de Rham cohomology of schemes, notes by I. Coates and O. Jussila, in *Dix exposés sur la cohomologie des schémas*, North-Holland.
- [8] A. Grothendieck, Un théorème sur les homomorphismes de schémas abéliens, *Invent. Math.* 2 (1966), 59–78.
- [9] P. A. Griffiths, Report on variation of Hodge structures, to appear.
- [10] M. Hakim, *Topos annelés et schémas relatifs*, Springer, to appear.
- [11] J.-I. Igusa, Abstract vanishing cycle theory, *Proc. Japan Acad.* 34, no. 9 (1958), 589–593.
- [12] R. Kiehl, Der Endlichkeitssatz für eigentliche Abbildungen in der nichtarchimedischen Funktionentheorie, *Invent. Math.* 2 (1967), 191–214.
- [13] R. Kiehl, further work on rigid analytic spaces and cohomology.
- [14] S. L. Kleiman, Algebraic cycles and the Weil conjectures, in *Dix exposés sur la cohomologie des schémas*, North-Holland.
- [15] S. Lang, *Abelian Varieties*, Interscience, 1959.
- [16] S. Lang and A. Néron, Rational points of abelian varieties over function fields, *Amer. J. Math.* 81 (1959), 95–118.
- [17] Yu. I. Manin, Theory of commutative formal groups, *Uspekhi Mat. Nauk* XVIII (1963), 3–90.
- [18] D. Mumford, *Geometric Invariant Theory*, Springer, 1965.
- [19] D. Mumford, Abstract theta functions, Advanced Science Seminar in Algebraic Geometry, Bowdoin College, 1967.
- [20] A. Néron, Modèles minimaux des variétés abéliennes sur les corps locaux et globaux, *Publ. Math. IHES* 21 (1964), 5–120.
- [21] M. Raynaud, Caractéristique d’Euler-Poincaré d’un faisceau et cohomologie des variétés abéliennes, in *Dix exposés sur la cohomologie des schémas*, North-Holland.
- [22] M. Raynaud, Modèles de Néron, *C. R. Acad. Sci. Paris* 262 (1966), 413–416.

- [23] M. Raynaud, Spécialisation du foncteur de Picard, *C. R. Acad. Sci. Paris* 264 (1967), 941–943 and 1001–1004.
- [24] M. Raynaud, Spécialisation du foncteur de Picard, to appear in *Publ. Math. IHES*.
- [25] M. Raynaud, Faisceaux amples sur les schémas en groupes et les espaces homogènes, thesis, Paris, 1968.
- [26] M. Raynaud, work on rigid analytic uniformization and Néron models.
- [27] M. Rosenlicht, Some basic theorems on algebraic groups, *Amer. J. Math.* 78 (1956), 401–443.
- [28] J.-P. Serre, Quelques propriétés des variétés abéliennes en caractéristique p , *Amer. J. Math.* 80 (1958), 715–739.
- [29] J.-P. Serre, *Corps locaux*, Hermann, Paris, 1962.
- [30] J.-P. Serre, Sur les corps locaux à corps résiduel algébriquement clos, *Bull. Soc. Math. France* 89 (1961), 105–154.
- [31] J.-P. Serre, *Abelian ℓ -adic Representations and Elliptic Curves*, Benjamin, 1968.
- [32] J.-P. Serre and J. Tate, Good reduction of abelian varieties, *Ann. of Math.* 88 (1968), 492–517.
- [33] J.-P. Serre and J. Tate, Elliptic curves and formal groups, Woods Hole Summer Institute notes, 1964.
- [34] J.-P. Serre, Morphismes universels et variété d’Albanese, Séminaire Chevalley 1958/59, Exposé 10.
- [35] J. Tate, WC-groups over p -adic fields, Séminaire Bourbaki no. 156, 1957.
- [36] J. Tate, p -divisible groups, Proceedings of a conference on local fields, Driebergen, 1966.
- [37] J. Tate, Rigid analytic spaces, mimeographed by IHES.